

Centered or Close?

Topic 5.4 · TODAY'S PROBLEM

Suppose a research team tests two rules for estimating a support proportion in a validation population where the true proportion is $p = 0.40$. For a random sample of 25 people, let X be the number who support the proposal.

Rule A reports $\hat{p}_A = X/25$. Rule B reports $\hat{p}_B = (X + 5)/35$. Exact summaries of the two sampling distributions are shown.

Lena: “A is unbiased, so it is always the more trustworthy estimator.”

Omar: “B varies less and is more likely to land within 0.06 of the truth, so its bias does not matter.”

Sampling distributions when $p = 0.40$

Rule	Mean	SD	Within 0.06
A	0.400	0.098	0.459
B	0.429	0.070	0.659

FIRST TAKE (5 min, on your sheet)

Whose recommendation do you trust more, Lena's or Omar's? Cite one entry from the table.

- DOK 1** (a) If one sample has $X = 9$ supporters, calculate the point estimate from Rule A and from Rule B. State the population parameter both rules estimate.
- DOK 2** (b) Use the table to classify each rule as biased or unbiased. For any biased rule, give the direction and amount of bias; then compare the rules' sampling variability.
- DOK 3** (c) Adjudicate both claims. For the goal of maximizing the chance that one estimate is within 0.06 of p , choose a rule using its center, spread, and within-0.06 probability. Explain why neither criterion means always better and what ignoring Rule B's bias could mislead a reader into believing.

Video + follow-along: u5_lesson4_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

